YASH LALA

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OVERVIEW

I'm a fourth-year PhD student at Yale, broadly interested in systems for efficient data-center scale computing. My research projects are generally themed around tradeoffs between memory capacity/bandwidth requirements and application performance. I approach these problems from a few different angles.

My primary specialty is in low-level operating systems and network stack optimization (via RDMA-based far memory systems on the CPU side, and MoE network performance modelling on the GPU side). I have 5+ years of kernel programming expertise, including large-scale projects such as reimplementing the memory-management subsystem in my most recent publication (SOSP '25). I have demonstrated expertise in developing high-throughput, low-latency kernel datapaths — such as a Linux-based system capable of migrating 200 Gib/s of memory at <270 ns per page per core. I'm deeply familiar with CPU microarchitecture and its interactions with the OS, and excel in principled, data-driven, bottleneck analyses.

Recently, I've also been tackling these problems from a high-level algorithm design/theory perspective. I feel that expertise in both worlds is necessary for effective algorithm-system codesign. I'm currently developing compact data structures for efficient nearest-neighbor queries in high-dimensional vector spaces. This project aims to accelerate RAG applications by marrying theory and practice: we provide formally proven runtime bounds, while maintaining strong empirical performance over billions of compressed RAG embeddings.

EDUCATION

Yale Ph.D. in Computer Science — Ongoing
UCLA B.S. in Computer Science — GPA: 3.782

PUBLICATIONS

Scalable Far Memory: Balancing Faults and Evictions — August 2023 - August 2025
31st Symposium for Operating Systems Principles, (SOSP 2025)

- Adapted the Linux Kernel to create a far-memory system capable of offloading excess application memory to arbitrary storage backends, for cluster-wide memory utilization improvements. Unlike prior work, this system scales to highly threaded, memory-intensive applications with minimal (near-ideal) application slowdown. Results are applicable to tiered memory systems and all far-memory backends (eg. NVMe, zswap, and RDMA).
- Rearchitected the Linux memory management subsystems to handle high-throughput page faulting and eviction. System achieves >200 Gib/s page migration throughput with only two CPU cores.
- Extensive low-level performance work around the OS-CPU interface, extending well into microarchitectural details. Gained expertise in: x86 IPI overheads and queuing characteristics; IOTLB-related virtualization overheads; TLB and page-walker coherence on x86; RDMA NIC control path throughput bottlenecks due to driver bugs; hypervisor-mediated IPI delivery costs; and lockless programming within the kernel.
- Developed principles for scalable memory migration. Results generalize to a Library-OS based re-implementation.
- Co-first author.

GRU4RecBE: Session Based Recommendations with Features — 2022
AAAI-22 Student Track

- Developed session-based recommendation system in PyTorch which extends the GRU4REC architecture with rich item features extracted from the pre-trained BERT architecture. Non-attentive model outperforms state-of-the-art session-based models over the MovieLens benchmark datasets.

PROFESSIONAL EXPERIENCE

NOVA Lab, Yale CS Department
PhD Student — *Advisor: Anurag Khandelwal*

- Broadly researching systems for efficient data-center scale computing. Recent work focuses on far-memory systems.
- Developed techniques for accelerating OS memory migration, aimed towards facilitating high-throughput RDMA-based far memory runtimes. Developed extensive knowledge of the the Linux kernel's memory management, RDMA, page placement, IPI, and swap subsystems. My particular focus was on optimizing high throughput, low-latency runtimes, such as memory migration systems and RDMA SmartNIC network stacks. Published in **SOSP '25**.
- Developing compact data structures for efficient range queries in high-dimensional vector spaces. Project is intended to accelerate floating-point vector databases for Retrieval-Augmented Generation (RAG) applications. Paper under submission.

NVIDIA
Systems Software Intern *Supervisor: Misha Smelyanskiy*

- Developed performance models for Mixture-of-Experts (MoE) communication phases for AI/ML models on NVIDIA's GB200 cluster hardware. Estimated ROI for future communication optimizations, and localized network performance bottlenecks in distributed systems of GPUs.
- Examined MoE token to expert mapping distributions to quantify the workload changes required for optimal hardware utilization. Explored algorithm-network codesign opportunities.
- Profiled existing GPU MoE library implementations (such as NCCL, DeepEP, etc). Optimized GPU MoE libraries for NVIDIA hardware by better utilizing tiered Multi-Node NVLink and Infiniband datacenter networks.
- Gained experience with LLM workload network traffic patterns and GPU performance debugging (particularly in GPU-initiated RDMA network communication).

SOLAR Lab, UCLA CS Department
Student Researcher *Supervisor: Harry Xu*

- Developed Linux kernel mechanisms for transparent far memory, with the goal of merging my changes upstream. The patchset extends the cpuset controller to allow per-cgroup control of active swap devices. The associated refactoring has positive implications for swap throughput, and makes it easy to manage frontswap-based remote memory systems. Patchset available at github.com/yashlala/canvas-linux.
- Developed a patchset to improve the Linux kernel's physical page allocation tail latencies by refilling the percpu low-order free page lists asynchronously using RCU.

CSSI Program, UCLA CS Department
Tutor Undergrad (TA)

- Taught introductory data science to high school students. Led discussion sections, prepared assignments, graded papers, and advised students.

Veritas Technologies LLC
SDE Intern

- Worked on NetBackup Flex, a platform for large-scale data consolidation and backup. Implemented automatic node discovery and cluster assimilation over datacenter networks. Primarily worked with Ansible, Docker, and various glue languages.

PROJECTS

NDN Multicast

- Extended routing protocols for NDN (Named Data Network) under Lixia Zhang. Extended NLSR (a link-state routing algorithm for NDN) to allow for efficient multicast delivery of NDN Interest packets. Student paper available at yashlala.com/nlsr-poster.pdf.

MISCELLANEOUS

Enjoys teaching: led student seminar series, volunteered with ACM TeachLA and as a docent for the Computer History Museum. Certified as a operator for the punched-card based IBM 1401.